

CHAITANYA KHARYAL

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SHORT RESEARCH STATEMENT

Through evolution and millions of years of testing in demanding and challenging environments, the human brain, and arguably complete human anatomy, has become very complex. So complex that it has become a challenge for humans even to understand our own abilities, let alone enabling computers to match our performance. Yet, this seemingly impossible task is the grand goal of many areas of research like Machine Learning, Deep Learning, Computer Vision, Robotics, and Reinforcement Learning, which is why they intrigue me as much as they do. My long term goal is to contribute to this grand goal of achieving human-level intelligence. By working towards this goal, I wish to understand more about our behaviours and decision making abilities while developing breakthrough intermediary technologies for the short term. I am particularly fascinated by the idea of combining existing classical model-based techniques with data-driven methods to make them more robust to unpredictable real-world scenarios.

EDUCATION

University of Alberta, Canada - <i>PhD in Computing Science</i>	2026-Present
University of Alberta, Canada - <i>Ms (Thes.) in Computing Science</i>	2024-2026 CGPA: 4.0
International Institute of Information Technology - Hyderabad, India - <i>B.Tech. Electronics and Communication Engineering (Hons.)</i> - <i>Honours: Robotics</i> <i>Advisor: Prof. Madhava Krishna</i> - <i>Awards and Honours:</i> • <i>Research Award</i> • <i>Dean's List</i>	2017-2021 CGPA: 9.04 5/8 semesters

PUBLICATIONS

Reward Learning Through Ranking Mean Squared Error - <i>Chaitanya Kharyal, Calarina Muslimani, Matthew E. Taylor</i> Developed Ranked Return Regression (R4), a rating-based reinforcement learning method that infers reward functions from human feedback. R4 uses a novel ranking MSE loss to learn from trajectory-rating pairs, providing formal guarantees and outperforming prior rating- and preference-based RL approaches on robotic locomotion benchmarks while requiring less feedback.	ICML'26, RLC'25 RLBrew workshop Website
GLIDE-RL: Grounded Language Instruction through DEmonstration in RL - <i>Chaitanya Kharyal, SaiKrishna Gottipati, Tanmay Kumar Sinha, Srijita Das, Matthew E. Taylor</i> One of the final frontiers in the development of complex human - AI collaborative systems is the ability of AI agents to comprehend the natural language and perform tasks accordingly. To this end, we present a novel algorithm, GroundedLanguage Instruction through DEmonstration in RL (GLIDE-RL) that introduces a teacher-instructor-student curriculum learning framework for training an RL agent capable of following natural language instructions that can generalize to previously unseen language instructions.	Extended Abstract accepted at AAMAS'24
Do As You Teach: A Multi-Teacher Approach to Self-Play in Deep Reinforcement Learning - <i>Chaitanya Kharyal, Tanmay Kumar Sinha, SaiKrishna Gottipati, Srijita Das, Matthew E. Taylor</i> We present a novel Multi-teacher approach to Self-play for sparse reward environments. Empirical results in Fetch-Reach and a novel driving simulator demonstrate that our proposed algorithm allows one agent (i.e., a teacher) to create a successful curriculum for another agent (i.e., the student). Surprisingly, results also show that training with multiple teachers actually helps the student learn faster.	NCAA, NeurIPS'22 DRL workshop, Extended abstract accepted at AAMAS'23
Spatial Relation Graph and Graph Convolutional Network for Object Goal Navigation - <i>D. A. Sasi Kiran, Chaitanya Kharyal, K. Anand, G. Kumar, N. Gireesh, S. Banerjee, R. dev Roychoudhury, M.</i>	CASE'22

We present a novel framework for the Object-goal navigation task that uses a history of walks to learn a Spatial Relational Graph (SRG) and a Graph Convolution network (GCN). While the SRG captures the inter region and object to region relationships, the GCN tries to capture valid trajectories to reach the goal. We also use a novel SRG based Bayesian approach to classify the regions in robots view just by using the visible objects. We report significant improvement in performance over relevant baselines.

Multi-Teacher Curriculum Design for Sparse Reward Environments

ALA workshop '22

- *Chaitanya Kharyal, Tanmay K. Sinha, Matthew E. Taylor*

Paper / Video

We present Adversarial Multi-Teacher Curriculum Design with Traces to learn efficiently in the sparse reward environments. This technique involves multiple independent teachers engaged in a game against a goal-conditioned student. The primary algorithmic novelty, relative to existing work, is engaging multiple teachers and using a behavior cloning loss. In addition, we also introduce a new sparse reward environment for simulated driving in PyBullet. Empirical results show the potential of our algorithm in this novel domain.

RP-VIO: Robust Plane-based Visual-Inertial Odometry for Dynamic Environments

IROS'21

- *Karnik Ram, Chaitanya Kharyal, Sudarshan S. Harithas, K. Madhava Krishna*

Paper / Code

We present a monocular visual-inertial odometry (VIO) system that uses only planar features and their induced homographies, during both initialization and sliding-window estimation, for increased robustness and accuracy in dynamic environments. We evaluate on diverse sequences, including our own highly-dynamic simulated dataset, and show significant improvement over a state-of-the-art monocular VIO algorithm in dynamic environments.

WORK EXPERIENCE

Microsoft, India

June 2021 - August 2024

Software Engineer

- Worked on improving the resiliency of Microsoft Azure services during Zonal failures. As a part of this team, tested several services during physical drills. Furthermore, ideated and created a platform to automate end-to-end drill management activities, using Azure functions, reducing the drill management time by 90%.
- Created a platform to run and oversee the life-cycle of quality validation workloads for Azure services and customers using Azure Kubernetes Services (AKS).
- Decreased the cost incurred for running the workloads by 3x by creating a historical data-based solution for scheduling workloads using AKS and Azure Functions.

Robotics Research Center, IIT - Hyderabad

May 2019 - May 2023

Researcher

- Worked on Object Goal Navigation task. Work accepted in CASE'22.
- Worked on improving the performance of Visual-Inertial Odometry (V-IO) algorithms in highly dynamic environments. Work accepted in IROS'21.

Matchday.ai

May 2019 - June 2019

Computer Vision and Machine Learning Intern

- Worked on ML and CV pipeline. Included ground segmentation of football fields, player classification and tracking, integration of different camera views etc.

IIT-Hyderabad, India

Teaching Assistant

- Embedded systems workshop (Approx 100 students)
- Linear Algebra (Approx 200 students)

Spring 2019

Monsoon 2019

THESIS

MSc Thesis. *Ranked Return Regression for Reward Learning in Reinforcement Learning*

RELEVANT COURSEWORK

Machine Learning and AI: Statistical Methods in AI, Probabilistic Graphical Models, Topics in Machine Learning (RL), **Robotics:** Mobile Robotics, Planning and Navigation **CS Theory:** Data Structures, Algorithms and Operating Systems, **Maths:** Probability and Random Processes, Linear Algebra and Graph Theory, Linear PDEs, **Other relevant courses:** Digital Image Processing, Computer System Organization, Signals and Systems

RELATED PROJECTS

Sparse Reward RL

- A self-lead research project trying to improve the learning of RL algorithms in sparse reward environments using asymmetric self play.

Gradient Evolution

Code / Blog

- Rediscovering the idea of gradient evolution presented in **The Evolutionary-Gradient-Search Procedure in Theory and Practice** from scratch.

Evolution Simulation

Code

- A simple simulation which simulates the evolution of simple organisms with simple traits such as speed, size, consciousness etc. using the Evolutionary Algorithm.

AWARDS, HONOURS AND ACCOMPLISHMENTS

- Winner of **Culture Award at Microsoft for Innovation**: Presented to be for developing a data-based solution that helped cut the cost incurred by the team by 3x.
- Winner of **ALA Cogment challenge at AAMAS**
- Winner of RL Hackathon by AICrowd
- Winner of JKPMSSS scholarship

REFERENCES

Matthew E. Taylor

University of Alberta

- Research Advisor
- matthew.e.taylor@ualberta.ca

Roopesh Rajamani

Microsoft

- Software Engineering Manager
- roopesh.rajamani@microsoft.com

K. Madhava Krishna

RRC, IIIT-H

- Research Advisor
- mkrishna@iiit.ac.in